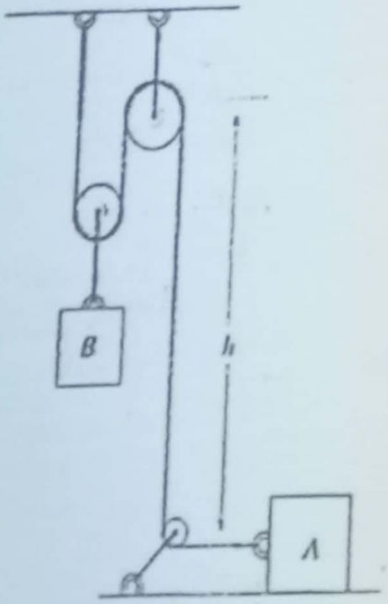
Relative Motion

Example:- Find the relation Between velocity and acceleration when Blocks A and B moves



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Datum

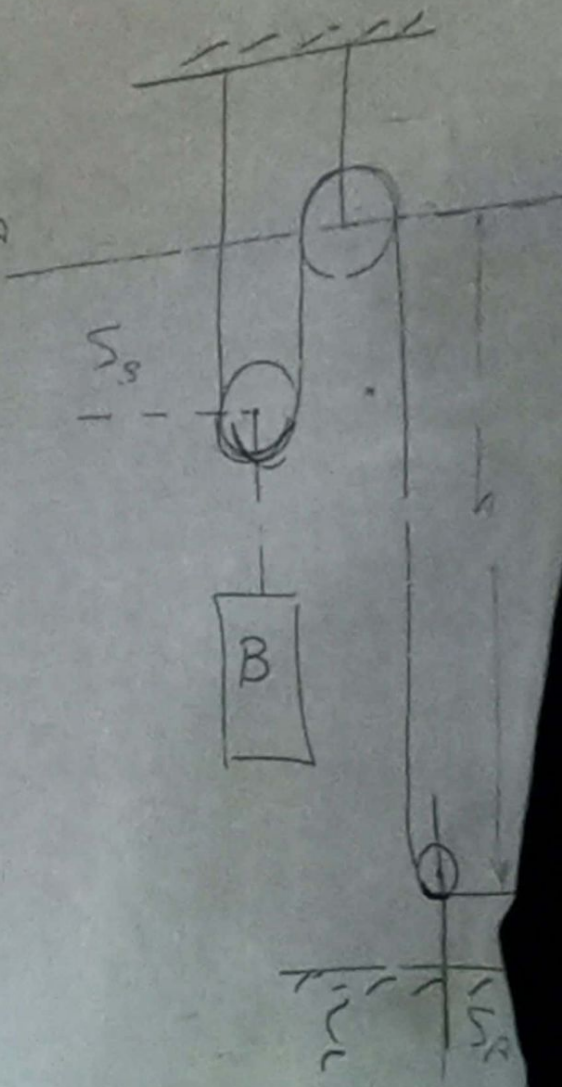
$$l = S_B + l_c + S_B + l_n + h + l_c + S_A$$

$$0 = \frac{dS_B}{dt} + \frac{dS_B}{dt} + \frac{dS_A}{dt}$$

$$0 = 2V_B + V_A$$

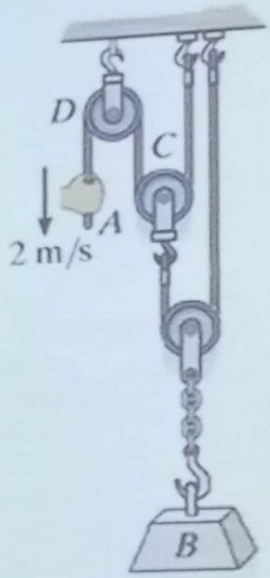
$$\boxed{V_A = -2V_B}$$

$$a_A = -2a_B$$



Relative Motion

Example: If the end of the cable at A is pulled down with a speed of 2 m/s, determine the speed at which block B rises.



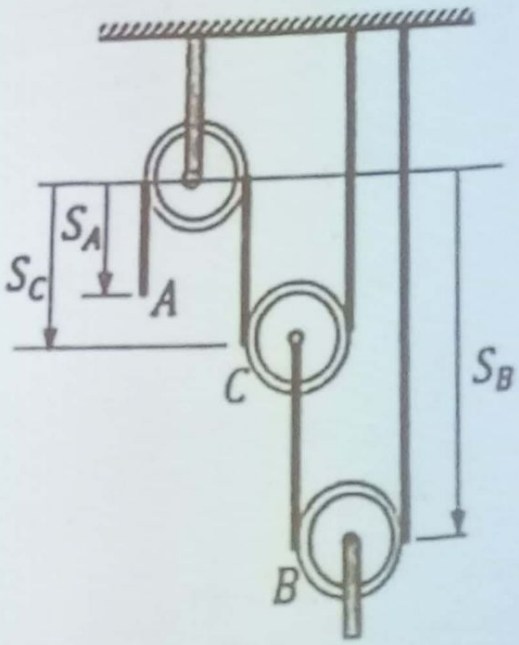
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Relative Motion

Step 1: Choose Datum

Datum is established at fixed pulley D

Step 2: The position of point A, block B and pulley C with respect to datum are SA , SB .and Sc. Since the system consists of two cords, two position coordinate equations can be developed.



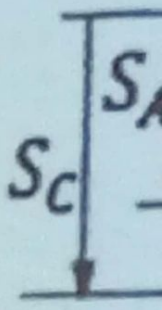
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Relative Motion

$$2S_C + S_A = l_1$$

$$S_B + (S_B - S_C) = l_2$$

$$S_A + 4S_B = l_1 + 2l_2$$



Time Derivative: Taking the time derivative of the above equation yields

$$v_A + 4v_B = 0$$

Since $V_1 = 2$ m/s

$$2 + 4v_B = 0$$

$$v_B = -\frac{2}{4} = -0.5$$

$$= 0.5 \text{ m/s} \uparrow$$

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$l_1 = S_A + S_C + S_C$

$l_1 = S_A + 2S_C \longrightarrow [1]$

$l_2 = (S_B - S_C) + S_B$

$l_2 = 2S_B - S_C \longrightarrow [2]$

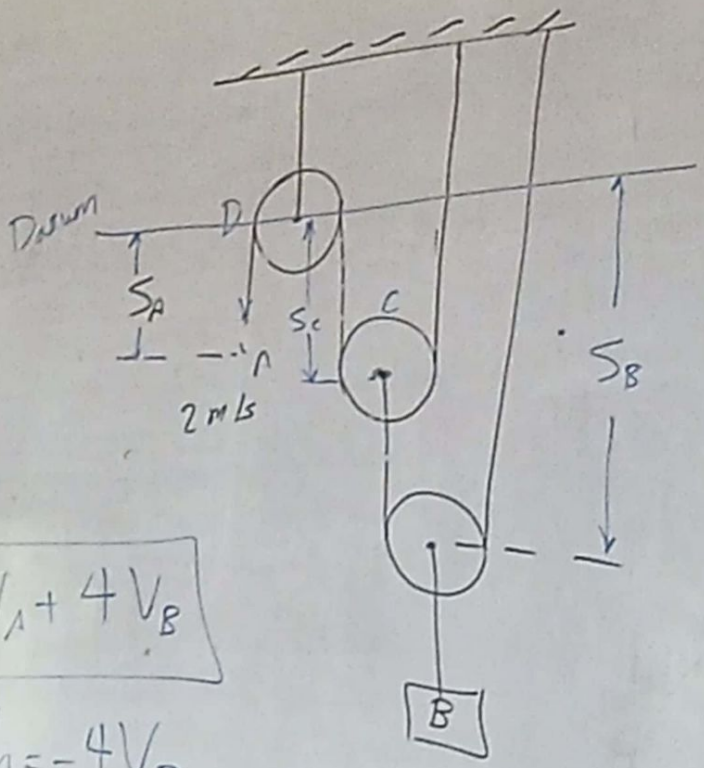
$0 = V_A + 2V_C$

$V_C = 2V_B$

$0 = 2V_B - V_C$

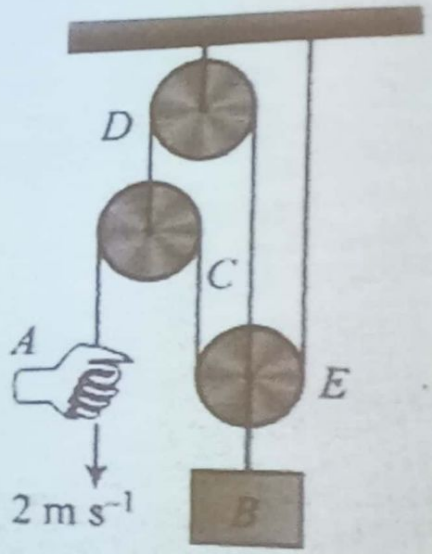
$0 = V_A + 4V_B$

$V_A = -4V_B$



Relative Motion

Example: Determine the speed of block B in Fig. if the end of the cord at A is pulled down with a speed of 2 m/s.



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Relative Motion

$$S_C + S_B = l_1$$

$$(S_A - S_C) + (S_B - S_C) + S_B = l_2$$

The time derivative of each equation gives

$$v_C + v_B = 0$$

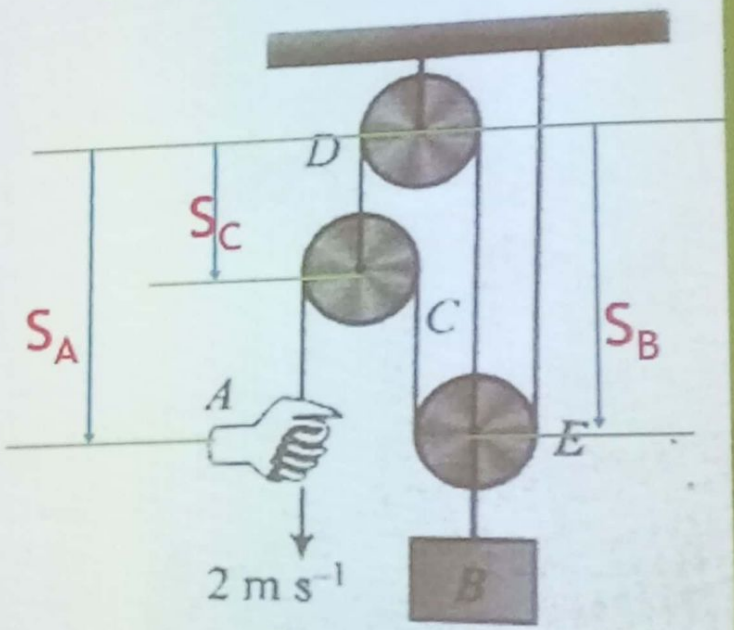
$$v_A - 2v_C + 2v_B = 0$$

Eliminating V_c, we obtain

$$v_C + 4v_B = 0$$

$$v_A = 2 \text{ m/s (downward)}$$

$$v_B = -0.5 \text{ m/s} = 0.5 \text{ m/s} \uparrow$$



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Relative Motion

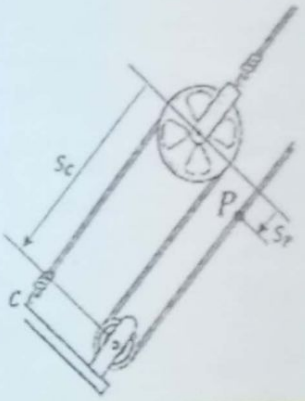
Solution

$$2S_C + (S_C - S_P) = l$$

$$2v_C + (v_C - v_P) = 0$$

$$3v_C - v_P = 0$$

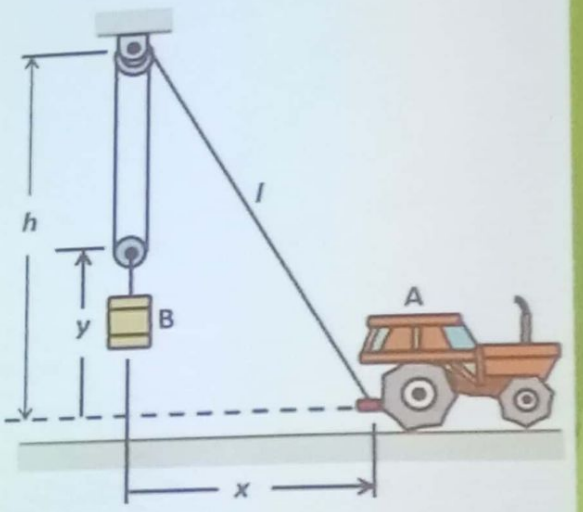
$$v_P = 3(-2) = -6 \text{ m/s}$$



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Relative Motion

Example: The tractor A is used to hoist the bale B. with the pulley arrangement shown. If A has a forward velocity V_A; determine an expression for the upward velocity V_B of the bale in terms of x.



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Relative Motion

The total constant length-of the cable is ·

$$L = 2(h - y) + l$$

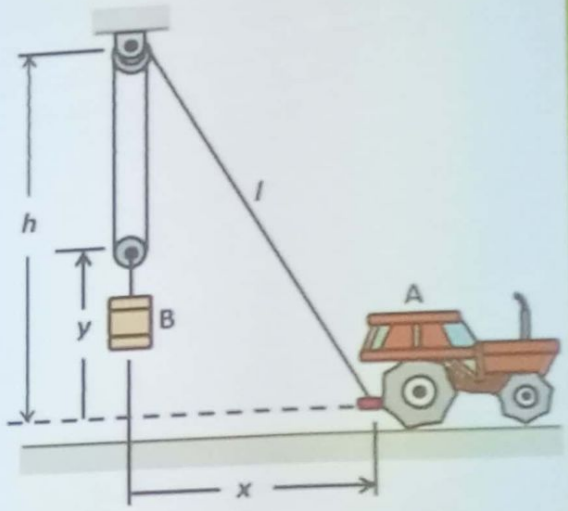
$$= 2(h - y) + \sqrt{h^2 + x^2}$$

Differentiation with time yields

$$0 = -2\frac{dy}{dt} + \frac{x\frac{dx}{dt}}{\sqrt{h^2 + x^2}}$$

Substituting $v_A = \frac{dx}{dt}$ and $v_B = \frac{dy}{dt}$ gives

$$v_B = \frac{1}{2}\frac{xv_A}{\sqrt{h^2 + x^2}}$$



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Force and Acceleration

1. Newton's Laws of Motion

Newton's second law of motion may be written in mathematical form as

$$F = ma$$

Equations of Motion for a Particle: Rectangular Coordinates

$$\sum F = ma$$

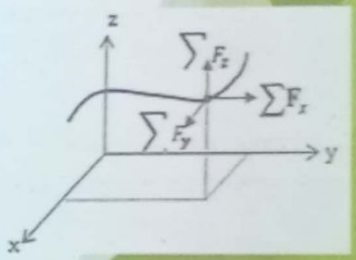
$$\sum F_x \mathbf{i} + \sum F_y \mathbf{j} + \sum F_z \mathbf{k} = m(a_x \mathbf{i} + a_y \mathbf{j} + a_z \mathbf{k})$$

the above equation can be represented in terms of the following three scalar equations:

$$\sum F_x = ma_x$$

$$\sum F_y = ma_y$$

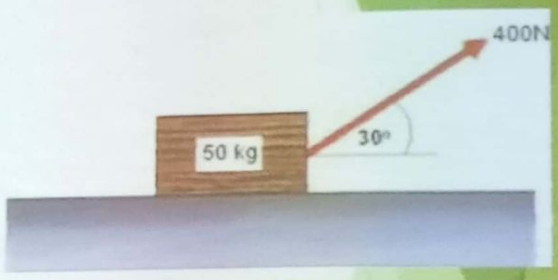
$$\sum F_z = ma_z$$



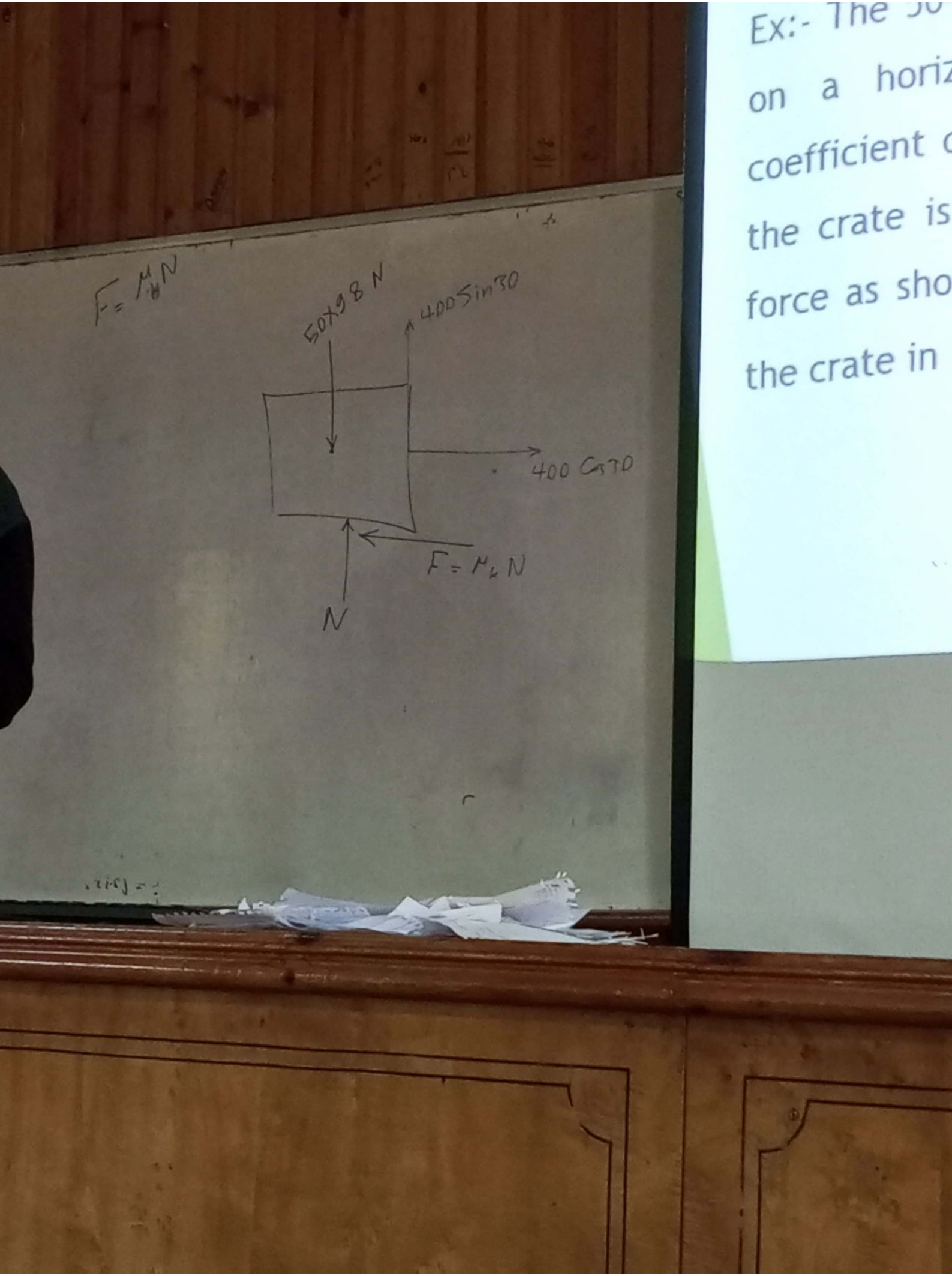
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Force and Acceleration

Ex:- The 50-kg crate shown in Figure rests on a horizontal plate for which the coefficient of kinetic friction is $\mu_k = 0.3$. If the crate is subjected to a 400-N towing force as shown, determine the velocity of the crate in 5 seconds starting from rest.



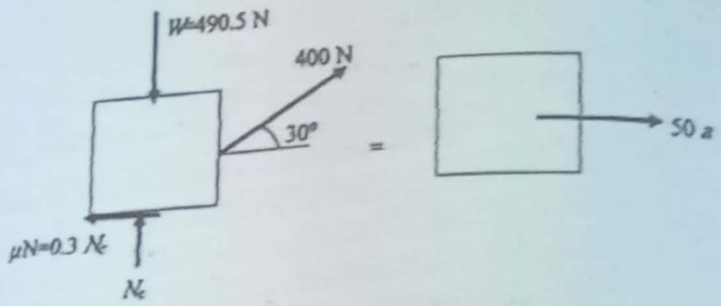
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Ex:- The
on a horiz
coefficient
the crate is
force as sho
the crate in
F = μ_R N
50×9.8 N
400 Sin 30
400 Cos 30
F = μ_L N
N

Force and Acceleration

Step 1: The weight of the crate is

$$W = mg = 50\ \text{kg}(9.81\ \text{m/s}^2) = 490.5\ \text{N}$$



Step 2: Applying the equations of motion

$$\sum F_x = ma_x; \qquad 400\cos 30^\circ - 0.3\,N_c = 50\,a$$

$$\sum F_y = ma_y; \qquad N_c - 490.5 + 400\sin 30^\circ = 0$$

Solving the two equations

$$N_c = 290.5\ \text{N}$$

$$a = 5.19\ \text{m/s}^2$$

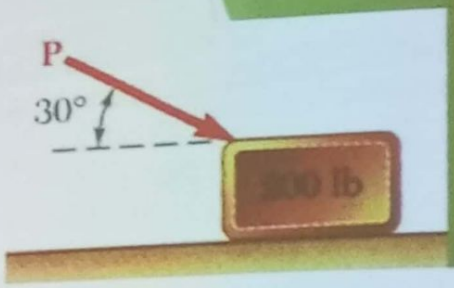
Step 3: Use the Equations of kinematics.

$$v = v_0 + at$$

$$= 0 + 5.19\,(5) = 26.0\ \text{m/s}.$$

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Force and Acceleration



SOLUTION:

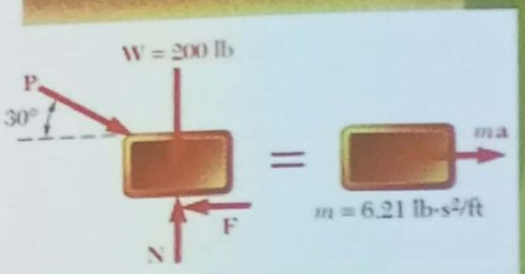
- Resolve the equation of motion for the block into two rectangular component equations.

$\sum F_x = ma$:

$$P\cos 30° - 0.25N = \left(6.21\,\text{lb}\cdot\text{s}^2/\text{ft}\right)\left(10\,\text{ft/s}^2\right)$$
$$= 62.1\,\text{lb}$$

$\sum F_y = 0$:

$$N - P\sin 30° - 200\,\text{lb} = 0$$



$$m = \frac{W}{g} = \frac{200\,\text{lb}}{32.2\,\text{ft/s}^2}$$
$$= 6.21\frac{\text{lb}\cdot\text{s}^2}{\text{ft}}$$

$$F = \mu_k N$$
$$= 0.25N$$